

MASTER'S COMPREHENSIVE EXAMINATION

Part I

Theory (90 minutes)

December 3 2005

✓ 1. An engineer observes independent lifetime measurements (= time to failure) on $n = 25$ circuit boards. The average of these 25 measurements is 1020. If the measurements come from an exponential distribution with mean θ , estimate θ and get an approximate 95% confidence interval for it.

2. A study of Georgia residents suggested that those who had worked (at some point) in a shipyard had a higher risk of dying from lung cancer. Death certificates and supporting records showed that 22% of those who died with lung cancer had worked at some point in a shipyard. By contrast, only 14% of those who died without lung cancer had worked in a shipyard. Suppose that .04% of all Georgia residents contract lung cancer. Find the percentage of lung cancer cases that will emerge among those Georgians now working (or who have worked) in a shipyard.

3. Let X_1 and X_2 be independent Poisson random variables with parameters $\lambda_1 = 1$ and $\lambda_2 = 2$, respectively.

(a) Find $P(X_1 = x \mid X_1 + X_2 = 3)$, for $x = 0, 2, 4$.

(b) Find the second moment of $X_1 X_2$.

4. Suppose X and Y are independent normally distributed random variables, both having variance = 2 and means μ_X and μ_Y , respectively. Test $H_0 : \mu_Y = \mu_X$ vs $H_1 : \mu_Y > \mu_X$ by rejecting H_0 if $Y - X \geq C$. Find C so that the test has significance level $\alpha = .05$ and find the power of the test when $\mu_Y = 3$, $\mu_X = 1$.